

DOM Events

Events are signals that something has occurred. (user inputs / actions)



<> index.html × # style.css JS app.js

<> index.html > html > body > button

```

1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>JS</title>
8      <link rel="stylesheet" href="style.css" />
9    </head>
10   <body>
11     <button>click me</button>
12     <script src="app.js"></script>
13   </body>
14 </html>
15

```



index.html • # style.css JS app.js

<> index.html > html > body > button

```
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>JS</title>
8      <link rel="stylesheet" href="style.css" />
9    </head>
10   <body>
11     <button onclick="console.log('button was clicked')">click me</button>
12     <script src="app.js"></script>
13   </body>
14 </html>
15
```



JS

× +

← → ↻

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

🔍 📄 ☆

APNA COLLEGE

click me

🔍 📄

Elements Console >>

⏮ ⏭ top 🔍 Filter

Default levels ▾ No Issues

button was clicked index.html:11

>



index.html × # style.css JS app.js

index.html > html > body > button

```

1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>JS</title>
8      <link rel="stylesheet" href="style.css" />
9    </head>
10   <body>
11     <button
12       | onclick="console.log('button was clicked'); console.log('apna college');"
13       | >
14       |   click me
15     </button>
16     <script src="app.js"></script>
17   </body>
18 </html>
19

```



DOM Events

onclick (when an element is clicked)

onmouseenter (when mouse enters an element)



< index.html

style.css

JS app.js



JS app.js > ...

```
1 let btn = document.querySelector("button");  
2 console.dir(btn);  
3 |
```



JS

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

click me!



APNA COLLEGE

Elements Console >>

top Filter

Default levels No Issues

```
onauxclick: null
onbeforecopy: null
onbeforecut: null
onbeforeinput: null
onbeforematch: null
onbeforepaste: null
onbeforetoggle: null
onbeforexrselect: null
onblur: null
oncancel: null
oncanplay: null
oncanplaythrough: null
onchange: null
onclick: null
onclose: null
oncontentvisibilityautostatechange: null
oncontextlost: null
oncontextmenu: null
oncontextrestored: null
oncopy: null
oncuechange: null
oncut: null
ondblclick: null
ondrag: null
ondragend: null
ondragenter: null
```

< index.html

style.css

JS app.js

JS app.js > onclick

```
1 let btn = document.querySelector("button");
2 console.dir(btn);
3
4 btn.onclick = function () {
5   console.log("button was clicked")
6 }
```



JS

×

+

←

→

↺

File

/Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

🔍

📄

☆

C:

APNA COLLEGE

⋮

click me!

🔍

📄

Elements

Console

⏏

🎬

🚫

top

👁

Filter

⚙

Default levels ▾

No Issues

▶ button

app.js:2

button was clicked

app.js:5

>

index.html

style.css

JS app.js

JS app.js > onclick

```
1 let btn = document.querySelector("button");
2 console.dir(btn);
3
4 btn.onclick = function () {
5   alert("button was clicked");
6 };
7
```

alert

ariaRoleDescription

SpeechRecognitionAlternative

function alert(message?: any): void

[MDN Reference](#)



<> index.html

style.css

JS app.js

JS app.js > ...

```
1 let btn = document.querySelector("button");
2 console.dir(btn);
3
4 // btn.onclick = function () {
5 //   alert("button was clicked");
6 // };
7
8 function sayHello() {
9   alert("Hello!");
10 }
11
12 btn.onclick = sayHello;
```

armankhan082019@gmail.com



This page says

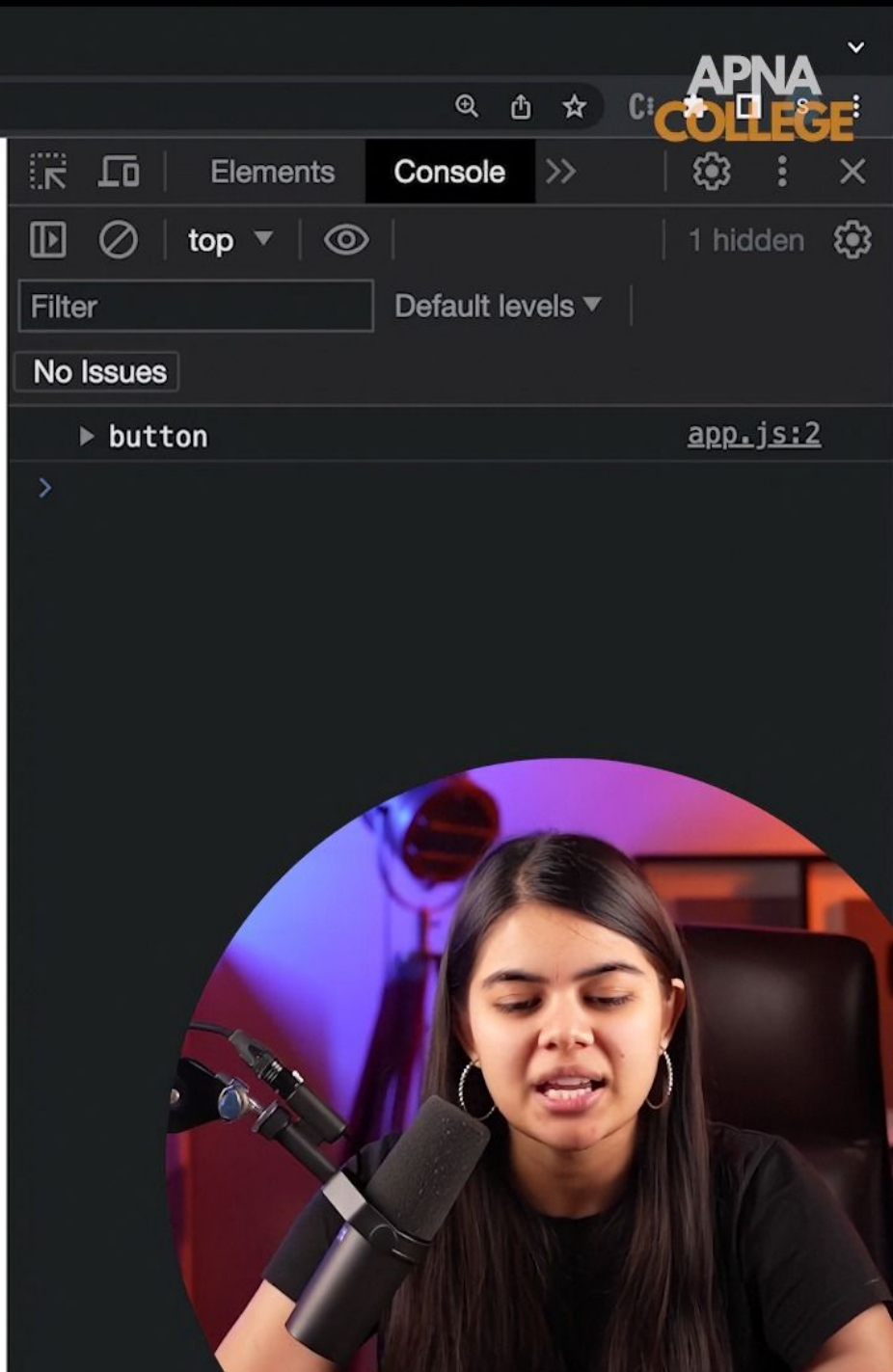
OK

  top  Filter 

► **button**

app.js:2





<> index.html

style.css

JS app.js



JS app.js > ...

```
1 let btn = document.querySelector("button");
2 console.dir(btn);
3
4 // btn.onclick = function () {
5 //   alert("button was clicked");
6 // };
7
8 function sayHello() {
9   alert("Hello!");
10 }
11
12 btn.onclick = sayHello;
13
```



04:52 / 08:56



<> index.html

style.css

JS app.js



JS app.js > ...

```
1 let btns = document.querySelectorAll("button");
2
3 for (btn of btns) {
4   btn.onclick = sayHello;
5 }
6
7 function sayHello() {
8   alert("Hello!");
9 }
10
```



JS

⌕

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

⌕ ⬆ ⬆

⌕ ⬆ ⬆

like me! like me 2! like me 3! like me 4!

This page says
Hello!

OK

Elements Console >>

top

Filter

Default levels

No Issues

>





index.html × # style.css JS app.js

index.html > html > body > script

```

1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>JS</title>
8      <link rel="stylesheet" href="style.css" />
9    </head>
10   <body>
11     <button>like me!</button>
12     <button>like me 2!</button>
13     <button>like me 3!</button>
14     <button>like me 4!</button>
15     <script src="app.js"></script>
16   </body>
17 </html>
18

```

Follow link (cmd + click)



DOM Events

onclick (when an element is clicked)

onmouseenter (when mouse enters an element)



<> index.html

style.css

JS app.js



JS app.js > onmouseenter

```

1  let btns = document.querySelectorAll("button");
2
3  for (btn of btns) {
4      btn.onclick = sayHello;
5      btn.onmouseenter = function () {
6          console.log("you entered a button");
7      };
8      console.dir(btn);
9  }
10
11  function sayHello() {
12      alert("Hello!");
13  }
14

```



JS

×

+

←

→

↻

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

🔍

📄

☆

C:📁

APNA COLLEGE

like me!

like me 2!

like me 3!

like me 4!

🔍

📄

Elements

Console

⏏

🔍

🚫

top

👁

Filter

⚙

Default levels ▾

No Issues

▶ button

app.js:8

▶ button

app.js:8

▶ button

app.js:8

▶ button

app.js:8

you entered a button

app.js:6

>

< index.html

style.css

JS app.js



JS app.js > onmouseenter

```
1 let btns = document.querySelectorAll("button");
2
3 for (btn of btns) {
4   btn.onclick = sayHello;
5   btn.onmouseenter = function () {
6     console.log("you entered a button");
7   };
8   console.dir(btn);
9 }
10
11 function sayHello() {
12   alert("Hello!");
13 }
14
```



Event Listener

addEventListener

element.addEventListener(*event*, *callback*)

```
btn.addEventListener("click", function () {  
  console.log("button clicked");  
});
```



<> index.html

style.css

JS app.js



JS app.js > ...

```
1 let btns = document.querySelectorAll("button");
2
3 for (btn of btns) {
4   btn.onclick = sayHello;
5   btn.onclick = sayName;
6 }
7
8 function sayHello() {
9   alert("Hello!");
10 }
11
12 function sayName() {
13   alert("Apna College");
14 }
15
```



Event Listener

addEventListener

click
drag
keyboard key

`element.addEventListener( event, callback )`

```
btn.addEventListener("click", function () {  
  console.log("button clicked");  
});
```



<> index.html

style.css

JS app.js

✕

JS app.js > ...

```
1 let btns = document.querySelectorAll("button");
2
3 for (btn of btns) {
4   // btn.onclick = sayHello;
5   // btn.onclick = sayName;
6
7   btn.addEventListener("click", sayHello);
8   btn.addEventListener("click", sayName);
9 }
10
11 function sayHello() {
12   alert("Hello!");
13 }
14
15 function sayName() {
16   alert("Apna College");
17 }
18
```

armankhan082020@gmail.com



References > Web APIs > Element

Filter

Document Object Model

Element

▼ Instance properties

- ariaAtomic
- ariaAutoComplete
- ariaBusy
- ariaChecked
- ariaColCount
- ariaColIndex
- ariaColSpan
- ariaCurrent
- ariaDescription
- ariaDisabled
- ariaExpanded
- ariaHasPopup

Fired when a key is released.

Mouse events

[auxclick](#)

Fired when a non-primary pointing device button (e.g., any mouse button other than the left button) has been pressed and released on an element.

[click](#)

Fired when a pointing device button (e.g., a mouse's primary button) is pressed and released on a single element.

[contextmenu](#)

Fired when the user attempts to open a context menu.

[dblclick](#)

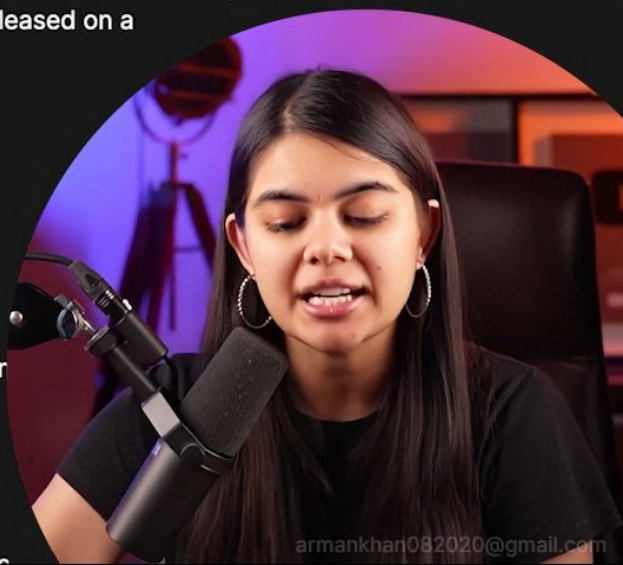
Fired when a pointing device button (e.g., a mouse's primary button) is clicked twice on an element.

[DOMActivate](#)

Occurs when an element is activated, for instance, through a mouse click or a keypress.

In this article

- Instance properties
- Instance methods
- Events**
- Specifications
- Browser compatibility



armankhan082020@gmail.com

<> index.html

style.css

JS app.js

JS app.js > ...

```

1  let btns = document.querySelectorAll("button");
2
3  for (btn of btns) {
4      // btn.onclick = sayHello;
5      // btn.onclick = sayName;
6
7      btn.addEventListener("click", sayHello);
8      btn.addEventListener("click", sayName);
9      btn.addEventListener("dblclick", sayName);
10 }
11
12 function sayHello() {
13     alert("Hello!");
14 }
15
16 function sayName() {
17     alert("Apna College");
18 }
19

```



<> index.html

style.css

JS app.js

×

JS app.js >  btn.addEventListener("dblclick") callback

```

1  let btns = document.querySelectorAll("button");
2
3  for (btn of btns) {
4      // btn.onclick = sayHello;
5      // btn.onclick = sayName;
6
7      btn.addEventListener("click", sayHello);
8      btn.addEventListener("click", sayName);
9      btn.addEventListener("dblclick", function () {
10         console.log("you double clicked me");
11     });
12 }
13
14 function sayHello() {
15     alert("Hello!");
16 }
17
18 function sayName() {
19     alert("Apna College");
20 }
21

```



JS

EventTarget: addEventListener

Event reference | MDN

Element - Web APIs | MDN

+

← → ↻

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

🔍 📄 ☆ C: APNA COLLEGE

like me! like me 2! like me 3! like me 4!

Elements Console >>

⏮ ⏭ top 🔍 Filter

Default levels ▾ No Issues

you double clicked me app.js:10

>

JS

EventTarget: addEventListener

Event reference | MDN

Element - Web APIs | MDN

+

developer.mozilla.org/en-US/docs/Web/API/Element#mouse_events

APNA COLLEGE

mdn web docs

References

Guides

Plus

Blog

Play

AI Help

Theme

Log in

Sign up for free

English (US)

References > Web APIs > Element

Filter

Document Object Model

Element

▼ Instance properties

ariaAtomic

ariaAutoComplete

ariaBusy

ariaChecked

ariaColCount

ariaColIndex

ariaColSpan

ariaCurrent

ariaDescription

ariaDisabled

ariaExpanded

ariaHasPopup

Fired when a pointing device button (e.g., a mouse's primary button) is clicked twice on a single element.

DOMActivate

Occurs when an element is activated, for instance, through a mouse click or a keypress.

mousedown

Fired when a pointing device button is pressed on an element.

mouseenter

Fired when a pointing device (usually a mouse) is moved over the element that has the listener attached.

mouseleave

Fired when the pointer of a pointing device (usually a mouse) is moved out of an element the listener attached to it.

mousemove

Fired when a pointing device (usually a mouse) is moved while over an element.

mouseout

In this article

Instance properties

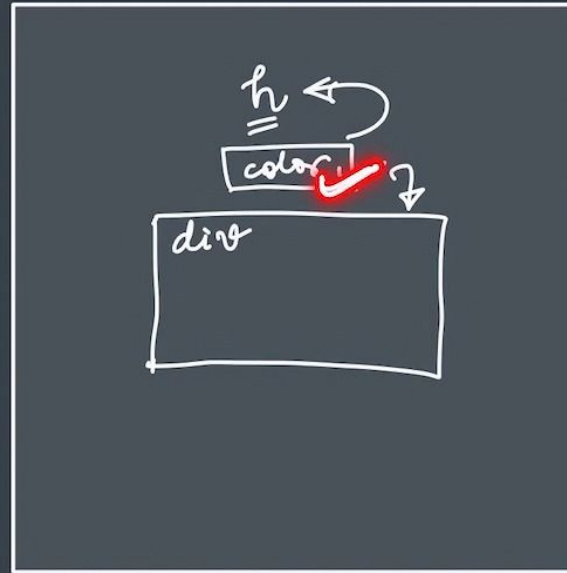
Instance methods

Events

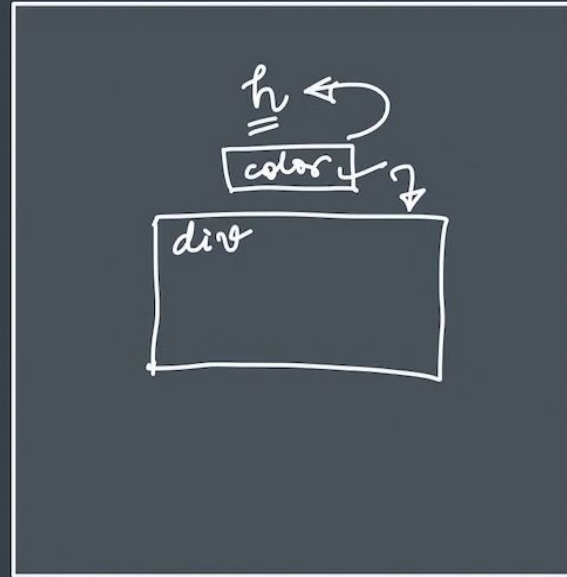
Specifications

Browser compatibility

Activity



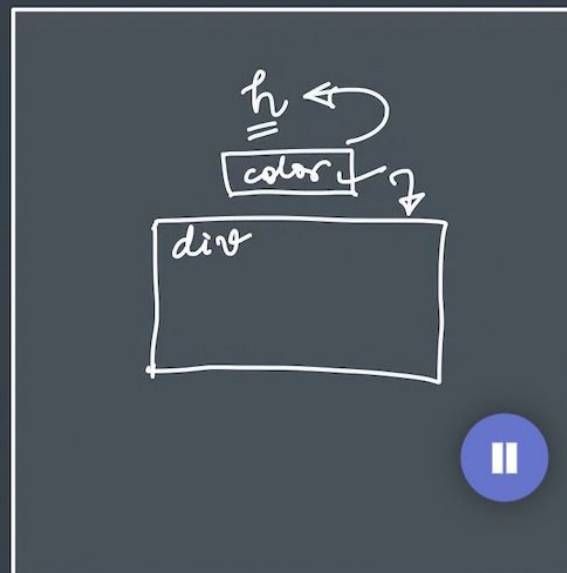
Activity



red R
green G
blue B



Activity



red R 0-255 ✓
green G 0-255 ✓
blue B 0-255 ✓

APNA
COLLEGE

JS



<> index.html

style.css

JS app.js

×

JS app.js > btn.addEventListener("click") callback

```

1  let btn = document.querySelector("button");
2
3  btn.addEventListener("click", function () {
4      let h3 = document.querySelector("h3");
5      let randomColor = getRandomColor();
6      h3.innerText = randomColor;
7
8      let div = document.querySelector("div");
9      div.style.backgroundColor = randomColor;
10
11     console.log("color updated");
12 });
13
14 function getRandomColor() {
15     let red = Math.floor(Math.random() * 255);
16     let green = Math.floor(Math.random() * 255);
17     let blue = Math.floor(Math.random() * 255);
18
19     let color = `rgb(${red}, ${green}, ${blue})`;
20     return color;
21 }
22

```



JS

× +

← → ↻

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

rgb(82, 109, 118)

Generate Color

This is your new color

APNA COLLEGE

Console >>

top

Filter

Default levels ▾

No Issues

color updated app.js:11

>

<> index.html

style.css

JS app.js



JS app.js > box.addEventListener("mouseenter") callback

```
1 let p = document.querySelector("p");
2
3 p.addEventListener("click", function () {
4   console.log("parah was clicked");
5 });
6
7 let box = document.querySelector(".box");
8 box.addEventListener("mouseenter", function () {
9   console.log("mouse inside div");
10 });
11
```



armankhan082020@gmail.com



02:09 / 02:43





The screenshot shows the Chrome DevTools Console. The 'Console' tab is selected. There are two log messages:

- Message 1: "parah was clicked" (logged from app.js:4)
- Message 2: "mouse inside box" (logged from app.js:9)



<> index.html

style.css

JS app.js X

```
JS app.js > btn.addEventListener("click") callback
1 let btn = document.querySelector("button");
2
3 btn.addEventListener("click", function () {
4   | console.log(this);
5   | });
6
```



JS

× +

← → ↺

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

🔍 📄 ☆ C: APNA COLLEGE

click me!

⏸

Elements Console >>

📄 🔍 top 🔍 Filter

Default levels ▾ No Issues

<button>click me!</button>

app.js:4

<button>click me!</button>

app.js:4

>

01:43 / 07:49

⏸ 🔊 🔍

smartsharad82@gmail.com

<> index.html

style.css

JS app.js X

```
JS app.js > btn.addEventListener("click") callback
1 let btn = document.querySelector("button");
2
3 btn.addEventListener("click", function () {
4   console.dir(this.innerText);
5 });
6
```



<> index.html

style.css

JS app.js X

```
JS app.js > btn.addEventListener("click") callback
1 let btn = document.querySelector("button");
2
3 btn.addEventListener("click", function () {
4   console.dir(this.innerText);
5   this.style.backgroundColor = "blue";
6 });
7
```



JS

× +

← → ↻

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

click me!

APNA COLLEGE

Elements Console >>

top Filter

Default levels No Issues

click me! app.js:4

>

<> index.html

style.css

JS app.js



JS app.js > h1.addEventListener("click") callback

```
1 let btn = document.querySelector("button");
2 let p = document.querySelector("p");
3 let h1 = document.querySelector("h1");
4 let h3 = document.querySelector("h3");
5
6 btn.addEventListener("click", function () {
7   console.dir(this.innerText);
8   this.style.backgroundColor = "blue";
9 });
10
11 p.addEventListener("click", function () {
12   console.dir(this.innerText);
13   this.style.backgroundColor = "blue";
14 });
15
16 h1.addEventListener("click", function () {
17   console.dir(this.innerText);
18   this.style.backgroundColor = "blue";
19 });
20
21 h3.addEventListener("click", function () {
22   console.dir(this.innerText);
23   this.style.backgroundColor = "blue";
24 });
25
```



JS

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

Use of 'this' in Event Listeners

Let's demo it on elements

This is a sample paragraph.

click me!

APNA COLLEGE

Elements Console

top Filter

Default levels No Issues

Use of 'this' in Event Listeners	app.js:17
Let's demo it on elements	app.js:22
This is a sample paragraph.	app.js:12
click me!	app.js:7

>



[illegible]

redundancy



<> index.html

style.css

JS app.js

X

JS app.js > ...

```
1 let btn = document.querySelector("button");
2 let p = document.querySelector("p");
3 let h1 = document.querySelector("h1");
4 let h3 = document.querySelector("h3");
5
6 function changeColor() {
7   console.dir(this.innerText);
8   this.style.backgroundColor = "blue";
9 }
10
11 btn.addEventListener("click", changeColor);
12
13 p.addEventListener("click", changeColor);
14
15 h1.addEventListener("click", changeColor);
16
17 h3.addEventListener("click", changeColor);
18
```



Use of 'this' in Event Listeners

Let's demo it on elements

This is a sample paragraph.

click me!



Use of 'this' in Event Listeners app.js:7

Let's demo it on elements app.js:7

This is a sample paragraph. app.js:7

>



Keyboard Events

JS



<> index.html

style.css

JS app.js



JS app.js > btn.addEventListener("click") callback

```
1 let btn = document.querySelector("button");
2 
3 btn.addEventListener("click", function(event) {
4   console.log(event);
5   console.log("button clicked");
6 });
7
```

armankhan082020@gmail.com



01:41 / 15:01



JS x +

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

click me!



APNA COLLEGE

Elements Console >>

top Filter

Default levels No Issues

```
app.js:4
  ▶ PointerEvent {isTrusted: true, pointerId: 1, width: 1, height: 1, pressure: 0, ...}
button clicked app.js:5
app.js:4
  ▶ PointerEvent {isTrusted: true, pointerId: 1, width: 1, height: 1, pressure: 0, ...}
button clicked app.js:5
app.js:9
  ▶ MouseEvent {isTrusted: true, screenX: 86, screenY: 114, clientX: 57, clientY: 23, ...}
button clicked app.js:10
>
```

<> index.html

style.css

JS app.js



JS app.js > ...

```
1 let btn = document.querySelector("button");
2
3 // btn.addEventListener("click", function (event) {
4 //   console.log(event);
5 //   console.log("button clicked");
6 // });
7
8 btn.addEventListener("dblclick", function (event) {
9   console.log(event);
10  console.log("button clicked");
11 });
12
```



JS

Event reference | MDN

Element - Web APIs | MDN

+

developer.mozilla.org/en-US/docs/Web/API/Element#keyboard_events

APNA COLLEGE

mdn web docs

References

Guides

Plus

Blog

Play

AI Help

Theme

Log in

Sign up for free

References > Web APIs > Element

English (US)

Filter

Document Object Model

Element

▼ Instance properties

ariaAtomic

ariaAutoComplete

ariaBusy

ariaChecked

ariaColCount

ariaColIndex

ariaColSpan

ariaCurrent

ariaDescription

ariaDisabled

ariaExpanded

ariaHasPopup

Sent to an **Element** when it transitions into or out of **fullscreen** mode.

fullscreenerror

Sent to an **Element** if an error occurs while attempting to switch it into or out of **fullscreen** mode.

Keyboard events

keydown

arman khan082020@gmail.com

Fired when a key is pressed.

keypress

Fired when a key that produces a character value is pressed down.

keyup

Fired when a key is released.

Mouse events

auxclick

Fired when a non-primary pointing device button (e.g., any mouse button other than the

In this article

Instance properties

Instance methods

Events

Specifications

Browser compatibility

<> index.html

style.css

JS app.js



JS app.js > inp.addEventListener("keydown") callback

```
1 let inp = document.querySelector("input");  
2  
3 inp.addEventListener("keydown", function () {  
4   console.log("key was pressed");  
5 });  
6
```

armankhan082020@gmail.com



04:44 / 15:01



JS

Event reference | MDN

Element - Web APIs | MDN

+

File

/Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

abcdjasbdjsdbfjbsd

click me!

Elements

Console

top

Filter

Default levels

No Issues

18

key was pressed

app.js:4

>

04:57 / 15:01

⚙️ 🔊 🔍



JS

Event reference | MDN

Element - Web APIs | MDN

+

← → ↻

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

🔍 📄 ☆

APNA COLLEGE

abcd

click me!

Elements

Console

top

Filter

Default levels

No Issues

32 key was pressed app.js:4

>

JS

Event reference | MDN

Element - Web APIs | MDN

+

← → ↻

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

🔍 📄 ☆

APNA COLLEGE

ab.

click me!

Elements Console >>

top 🔍 Filter

Default levels ▾ No Issues

64 key was pressed app.js:4

>

<> index.html

style.css

JS app.js

```
JS app.js > inp.addEventListener("keyup") callback
1 let inp = document.querySelector("input");
2
3 // inp.addEventListener("keydown", function () {
4 //   console.log("key was pressed");
5 // });
6
7 inp.addEventListener("keyup", function () {
8   console.log("key was released");
9 });
10
```

armankhan082020@gmail.com



JS

Event reference | MDN

Element - Web APIs | MDN

+

← → ↻

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

🔍 📄 ☆

APNA COLLEGE

abcddc

click me!

Elements

Console

Filter

Default levels ▾

No Issues

5 key was released app.js:8

>

<> index.html

style.css

JS app.js

```
JS app.js > inp.addEventListener("keydown") callback
1   let inp = document.querySelector("input");
2
3   inp.addEventListener("keydown", function (event) {
4       console.log(event)
5       console.log("key was pressed");
6   });
7
8   // inp.addEventListener("keyup", function () {
9       //   console.log("key was released");
10  // });
11
```

armankhan082020@gmail.com



JS

Event reference | MDN

Element - Web APIs | MDN

+

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

a

click me!

Elements

Console

top

Filter

Default levels

No Issues

app.js:4

KeyboardEvent {isTrusted: true, key: 'a', code: 'KeyA', location: 0, ctrlKey: false, ...}

key was pressed

app.js:5

06:20 / 15:01

APNA COLLEGE

06:20 / 15:01

JS

Event reference | MDN

Element - Web APIs | MDN

+

File

/Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

ab ;

click me!

Elements

Console

top

Filter

Default levels

No Issues

e: 'KeyA', location: 0, ctrlKey: false, ...

key was pressed

app.js:5

app.js:4

KeyboardEvent {isTrusted: true, key: 'b', code: 'KeyB', location: 0, ctrlKey: false, ...}

key was pressed

app.js:5

app.js:4

KeyboardEvent {isTrusted: true, key: ' ', code: 'Space', location: 0, ctrlKey: false, ...}

key was pressed

app.js:5

app.js:4

KeyboardEvent {isTrusted: true, key: ';', code: 'Semicolon', location: 0, ctrlKey: false, ...}

isTrusted:

altKey: f

bubbles:

cancelBubble:

cancelPropagation:

charCode:

code:

composed:

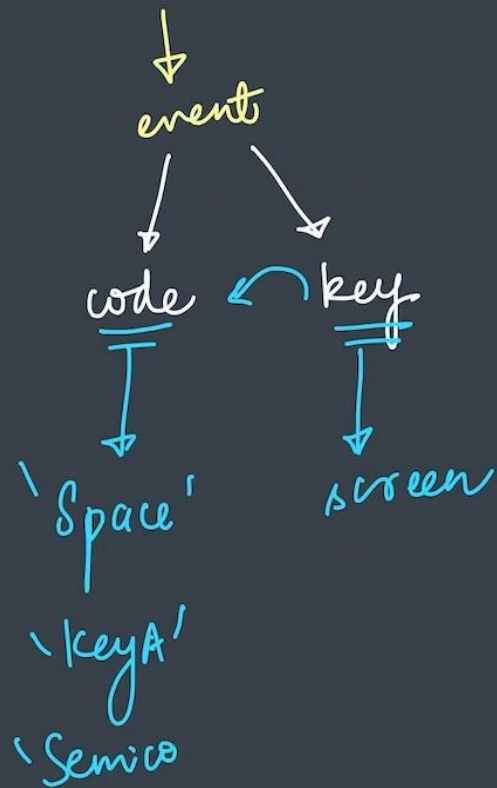
ctrlKey:

currentTarget:

defaultPrevented:

detail:

Keyboard Events



a
b
 ;
 "a"
 "b"
 " "
 ";"



JS

Event reference | MDN

Element - Web APIs | MDN

+

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

az/

click me!

APNA COLLEGE

Elements

Console

top

Filter

Default levels

No Issues

key = a

code = KeyA

key was pressed

key = z

code = KeyZ

key was pressed

key = Enter

code = Enter

key was pressed

key = /

code = Slash

key was pressed

>

09:24 / 15:01

09:24 / 15:01

09:24 / 15:01



JS

Event reference | MDN

Element - Web APIs | MDN

+

File

/Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

az/?

click me!

APNA COLLEGE

Elements

Console

top

Filter

Default levels

No Issues

key = a

app.js:4

code = KeyA

app.js:5

key was pressed

app.js:6

key = z

app.js:4

code = KeyZ

app.js:5

key was pressed

app.js:6

key = Enter

app.js:4

code = Enter

app.js:5

key was pressed

app.js:6

key = /

app.js:4

code = Slash

app.js:5

key was pressed

app.js:6

key = Shift

app.js:4

code = ShiftRight

app.js:5

key was pressed

app.js:6

key = ?

app.js:4

code = Slash

app.js:5

key was pressed

app.js:6

>

Elements Console >>

top ▾ Filter

Default levels ▾ No Issues

key = `	app.js:4
code = Backquote	app.js:5
key was pressed	app.js:6

>



JS

Event reference | MDN

Element - Web APIs | MDN

+

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

type something

click me!

Elements

Console

top

Filter

Default levels

No Issues

key = ArrowUp

code = ArrowUp

key was pressed

key = ArrowDown

code = ArrowDown

key was pressed

key = ArrowLeft

code = ArrowLeft

key was pressed

key = ArrowRight

code = ArrowRight

key was pressed

app.js:4

app.js:5

app.js:6

app.js:4

app.js:5

app.js:6

app.js:4

app.js:5

app.js:6

app.js:4

app.js:5

app.js:6



<> index.html

style.css

JS app.js

X

JS app.js > ...

```
1 let inp = document.querySelector("input");
2
3 inp.addEventListener("keydown", function (event) {
4   console.log("key = ", event.key);
5   console.log("code = ", event.code);
6   console.log("key was pressed");
7 });
8
9 // inp.addEventListener("keyup", function () {
10 //   console.log("key was released");
11 // });
12
```



<> index.html

style.css

JS app.js

X

JS app.js > inp.addEventListener("keydown") callback

```
1 let inp = document.querySelector("input");
2
3 inp.addEventListener("keydown", function (event) {
4   console.log("code = ", event.code); //ArrowUp, ArrowDown, ArrowLeft, ArrowRight
5   if (event.code == "ArrowUp") {
6     console.log("character moves forward");
7   } else if (event.code == "ArrowDown") {
8     console.log("character moves backward");
9   } else if (event.code == "ArrowLeft") {
10    console.log("character moves left");
11  } else if (event.code == "ArrowRight") {
12    console.log("character moves right");
13  }
14 });
15
```



JS

Event reference | MDN

Element - Web APIs | MDN

+

← → ↻

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

🔍 📄 ☆

APNA COLLEGE

move you character

click me!

Elements Console >>

🔍 top 🔍 Filter

Default levels ▾ No Issues

code = ArrowUp

app.js:4

character moves forward

app.js:6

code = ArrowDown

app.js:4

character moves backward

app.js:8

code = ArrowLeft

app.js:4

character moves left

app.js:10

>

<> index.html

style.css

JS app.js



JS app.js > inp.addEventListener("keydown") callback

```
1 let inp = document.querySelector("input");
2
3 inp.addEventListener("keydown", function (event) {
4   console.log("code = ", event.code); //ArrowUp(U), ArrowDown(D), ArrowLeft(L), ArrowRight(R)
5   if (event.code == "KeyU") {
6     console.log("character moves up");
7   } else if (event.code == "KeyD") {
8     console.log("character moves down");
9   } else if (event.code == "KeyL") {
10    console.log("character moves left");
11  } else if (event.code == "KeyR") {
12    console.log("character moves right");
13  }
14 });
15
```



JS

Event reference | MDN

Element - Web APIs | MDN

+

← → ↻ ⓘ File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

ulldr

click me!

APNA COLLEGE

Elements Console >>

top Filter

Default levels No Issues

code = KeyU app.js:4

character moves up app.js:6

code = KeyL app.js:4

character moves left app.js:10

code = KeyL app.js:4

character moves left app.js:10

code = KeyL app.js:4

character moves left app.js:10

code = KeyD app.js:4

character moves down app.js:8

code = KeyR app.js:4

character moves right app.js:12

>

<> index.html • # style.css JS app.js

<> index.html > html > body > form > button

```
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>JS</title>
8      <link rel="stylesheet" href="style.css" />
9    </head>
10   <body>
11     <form action="/action">
12       <input placeholder="username">
13       <button>Register</button>
14     </form>
15     <script src="app.js"></script>
16   </body>
17 </html>
18
```





<> index.html

style.css

JS app.js X

JS app.js > form.addEventListener("submit") callback

```
1 let form = document.querySelector("form");
2
3 form.addEventListener("submit", function () {
4   console.log("form submitted");
5 });
6
```

armankhan082020@gmail.com



01:38 / 04:10

<> index.html

style.css

JS app.js



JS app.js > ...

```
1 let form = document.querySelector("form");
2 
3 form.addEventListener("submit", function () {
4   alert("form submitted");
5 });
6
```

armankhan082020@gmail.com



JS

× +

← → ↺

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

🔍 📄 ☆

APNA COLLEGE

⚙️ ⋮ ✕

abc

Register

This page says

form submitted

OK

Elements

Console

top

Filter

Alt levels

No Issues

amakhya68202@gmail.com

⏸



⏸

02:15 / 04:10

⚙️ 🔊 🔍

Form Events

event.preventDefault()

armankhan082020@gmail.com



<> index.html

style.css

JS app.js



JS app.js > ...

```
1 let form = document.querySelector("form");
2 
3 form.addEventListener("submit" function (event) {
4   event.preventDefault();
5   alert("form submitted");
6 });
7
```

armankhan082020@gmail.com



JS x +

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

apnacollege Register

APNA COLLEGE

Elements Console >>

top Filter

Default levels No Issues

▶ input	app.js:7
abcd	app.js:8
▶ input	app.js:7
abcdegh	app.js:8
▶ input	app.js:7
apnacollege	app.js:8

>

03:05 / 09:35

<> index.html

style.css

JS app.js

JS app.js >  form.addEventListener("submit") callback

```
1 let form = document.querySelector("form");
2 
3 form.addEventListener("submit", function (event) {
4   event.preventDefault();
5
6   let inp = document.querySelector("input");
7   console.dir(inp);
8   console.log(inp.value);
9 });
10
```



<> index.html ● # style.css JS app.js

<> index.html > html > body > form > input#user

```
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>JS</title>
8      <link rel="stylesheet" href="style.css" />
9    </head>
10   <body>
11     <form action="/action">
12       <input placeholder="username" type="text" id="user"/>
13       <input placeholder="password" type="password">
14       <button>Register</button>
15     </form>
16     <script src="app.js"></script>
17   </body>
18 </html>
19
```



<> index.html

style.css

JS app.js

JS app.js > form.addEventListener("submit") callback

```
1 let form = document.querySelector("form");
2
3 form.addEventListener("submit", function (event) {
4   event.preventDefault();
5
6   let user = document.querySelector("#user");
7   let pass = document.querySelector("#pass");
8
9
10 });
11
```



<> index.html

style.css

JS app.js

×

JS app.js > form.addEventListener("submit") callback

```
1 let form = document.querySelector("form");
2
3 form.addEventListener("submit", function (event) {
4   event.preventDefault();
5
6   let user = document.querySelector("#user");
7   let pass = document.querySelector("#pass");
8
9   console.log(user.value);
10  console.log(pass.value);
11 });
12
```

[x] pass
[x] PaymentResponse
[x] PermissionStatus
[x] SVGAnimatedPreserveAspectRatio
[x] DOCUMENT_POSITION_CONTAINS
[x] PerformanceObserverEntryList
[x] DOCUMENT_POSITION_IMPLEMENTATION_SPECIFIC

let pass: Element | null

×



JS

× +

← → ↻

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

shradha

....

Register

Elements Console >>

top Filter

Default levels No Issues

shradha app.js:9

abcd app.js:10

>





<> index.html

style.css

JS app.js

JS app.js > form.addEventListener("submit") callback

```
1 let form = document.querySelector("form");
2
3 form.addEventListener("submit", function (event) {
4   event.preventDefault();
5
6   let user = document.querySelector("#user");
7   let pass = document.querySelector("#pass");
8
9   console.log(user.value);
10  console.log(pass.value);
11
12  alert(`Hi ${user.value}, your password is set to ${pass.value}`);
13 });
14
```

alert(message?: any): void

[MDN Reference](#)



JS

× +

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

apnacollege

....

Re

This page says
Hi apnacollege, your password is set to abcd

OK

Elements

Console

top

Filter

Alt levels

No Issues

apnacollege

app.js:9

abcd

app.js:10

>

⏸



armankhan082020@gmail.com

⏸

06:37 / 09:35

⚙️ 🔊 🔍

<> index.html × # style.css JS app.js

<> index.html > html > body > form > input#user

```

1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>JS</title>
8      <link rel="stylesheet" href="style.css" />
9    </head>
10   <body>
11     <form action="/action">
12       <input placeholder="username" type="text" id="user" />
13       <input placeholder="password" type="password" id="pass" />
14       <button>Register</button>
15     </form>
16     <script src="app.js"></script>
17   </body>
18 </html>
19

```



<> index.html

style.css

JS app.js

×

JS app.js > form.addEventListener("submit") callback > [e] user

```

1  let form = document.querySelector("form");
2
3  form.addEventListener("submit", function (event) {
4      event.preventDefault();
5
6      let user = document.querySelector("#user");
7      let pass = document.querySelector("#pass");
8
9      console.log(user.value);
10     console.log(pass.value);
11
12     alert(`Hi ${user.value}, your password is set to ${pass.value}`);
13 });
14

```



<> index.html

style.css

JS app.js

×

JS app.js > form.addEventListener("submit") callback

```

1  let form = document.querySelector("form");
2
3  form.addEventListener("submit", function (event) {
4      event.preventDefault();
5      console.dir(form);
6
7      // let user = document.querySelector("#user");
8      // let pass = document.querySelector("#pass");
9
10     // console.log(user.value);
11     // console.log(pass.value);
12
13     // alert(`Hi ${user.value}, your password is set to ${pass.value}`);
14 });
15

```



ocd

...

Register

button 65.65 × 21



Elements Console >>

top Filter

Default levels No Issues

app.js:5

```
▼ form ⓘ  
  ▶ 0: input#user  
  ▶ 1: input#pass  
  ▶ 2: button  
    acceptcharset: ""  
    accessKey: ""  
    action: "file:///action"  
    ariaAtomic: null  
    ariaAutoComplete: null  
    ariaBrailleLabel: null  
    ariaBrailleRoleDescription: null  
    ariaBusy: null  
    ariaChecked: null  
    ariaColCount: null  
    ariaColIndex: null  
    ariaColSpan: null  
    ariaCurrent: null  
    ariaDefault: null  
    ariaDisabled: null  
    ariaExpanded: null  
    ariaHasPopup: null  
    ariaHidden: null  
    ariaInvalid: null  
    ariaKeyShortcuts: null  
    ariaLabel: null
```



JS

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

abcd

...

Register

Elements

Console

top

Filter

Default levels

No Issues

```
clientHeight: 21
clientLeft: 0
clientTop: 0
clientWidth: 575
contentEditable: "inherit"
dataset: DOMStringMap {}
dir: ""
draggable: false
elementTiming: ""
elements: HTMLFormControlsCollection(3) [input, ..., button]
encoding: "application/x-www-form-urlencoded"
enctype: "application/x-www-form-urlencoded"
enterKeyHint: ""
firstChild: text
firstElementChild: input#user
hidden: false
id: ""
inert: false
innerHTML: "\n      <input placeholder=\"user\"
innerText: "  Register"
inputMode: ""
isConnected: true
isContentEditable: false
lang: ""
lastChild: text
lastElementChild: button
```

armankhan082020@gmail.com

07:46 / 09:35

JS

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

abcd

...

Register

APNA COLLEGE

Elements Console

top Filter

Default levels No Issues

```
clientHeight: 21
clientLeft: 0
clientTop: 0
clientWidth: 575
contentEditable: "inherit"
▶ dataset: DOMStringMap {}
dir: ""
draggable: false
elementTiming: ""
▼ elements: HTMLFormControlsCollection(3)
  ▶ 0: input#user
  ▶ 1: input#pass
  ▶ 2: button
  ▶ pass: input#pass
  ▶ user: input#user
    length: 3
  ▶ [[Prototype]]: HTMLFormControlsCollection
encoding: "application/x-www-form-urlencoded"
enctype: "application/x-www-form-urlencoded"
enterKeyHint: ""
▶ firstChild: text
▶ firstElementChild: input#user
hidden: false
id: ""
inert: false
innerHTML: "\n    <input placeholder=\"user
```

armankhan082020@gmail.com

07:51 / 09:35

JS

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

abcd ... Register



APNA COLLEGE

Elements Console

top Filter

Default levels No Issues

```
> form
> form.elements
< HTMLFormControlsCollection(3) [input#user,
  > input#pass, button, user: input#user, pass:
    input#pass]
> form.elements[0];
< <input placeholder="username" type="text" id=
  "user">
> form.elements[1];
< <input placeholder="password" type="password"
  id="pass">
> |
```


<> index.html

style.css

JS app.js

JS app.js > form.addEventListener("submit") callback

```

1  let form = document.querySelector("form");
2
3  form.addEventListener("submit", function (event) {
4      event.preventDefault();
5      console.dir(form);
6
7      let user = this.elements[0];
8      let pass = document.querySelector("#pass");
9
10     console.log(user.value);
11     console.log(pass.value);
12
13     alert(`Hi ${user.value}, your password is set to ${pass.value}`);
14 });
15

```



<> index.html

style.css

JS app.js



JS app.js > form.addEventListener("submit") callback

```
1 let form = document.querySelector("form");
2
3 form.addEventListener("submit", function (event) {
4   event.preventDefault();
5   console.dir(form);
6   //
7   let user = this.elements[0].value;
8   let pass = this.elements[1].value;
9
10  console.log(user.value);
11  console.log(pass.value);
12
13  alert(`Hi ${user.value}, your password is set to ${pass.value}`);
14 });
15
```



armankhan082020@gmail.com

JS

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

abcd

....

Re

This page says

Hi abcd, your password is set to 1234

OK

Elements Console

top Filter

ilt levels No Issues

form app.js:5

abcd app.js:10

1234 app.js:11

>

09:25 / 09:35

More Events

change event

The change event occurs when the value of an element has been changed (only works on <input>, <textarea> and <select> elements).

input event

armankhan082020@gmail.com

The input event fires when the value of an <input> , <select> , or <textarea> element has been changed.



<> index.html

style.css

JS app.js

×

JS app.js > user.addEventListener("change") callback

```

1  // let form = document.querySelector("form");
2
3  // form.addEventListener("submit", function (event) {
4  //   event.preventDefault();
5  //   console.dir(form);
6
7  //   let user = this.elements[0]; //form.elements[0]
8  //   let pass = this.elements[1];
9
10 //   console.log(user.value);
11 //   console.log(pass.value);
12
13 //   alert(`Hi ${user.value}, your password is set to ${pass.value}`);
14 // });
15
16 let user = document.querySelector("#user");
17
18 user.addEventListener("change", function () {
19   console.log("input changed");
20   console.log("final value = ", user.value);
21 });
22

```



abc password Register

makken082020@gmail.com

Elements Console >> Filter

Default levels ▾ No Issues

```
input changed app.js:10  
final value = abc app.js:11  
>
```



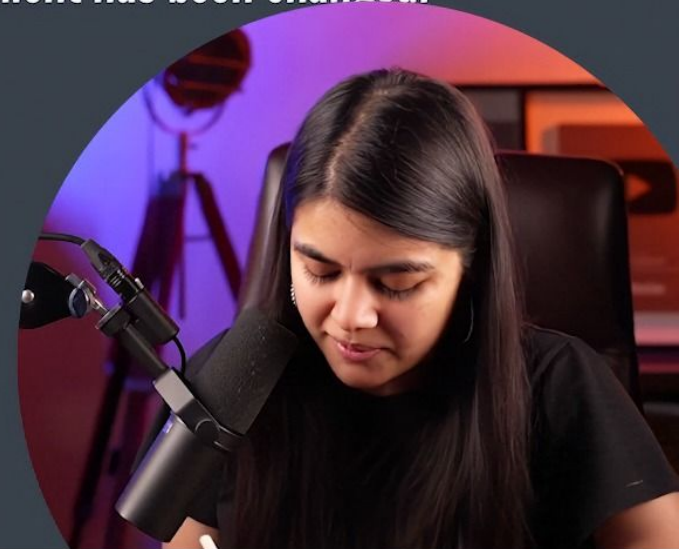
More Events

change event

The change event occurs when the value of an element has been changed (only works on `<input>`, `<textarea>` and `<select>` elements).

armankhan0 **input** event

The input event fires when the value of an `<input>` , `<select>` , or `<textarea>` element has been changed.



JS x +

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

abcde password Register

||



armankhan082020@gmail.com

APNA COLLEGE

Elements Console >>

top Filter

Default levels No Issues

input event	app.js:15
final value = a	app.js:16
input event	app.js:15
final value = ab	app.js:16
input event	app.js:15
final value = abc	app.js:16
input event	app.js:15
final value = abcd	app.js:16
input event	app.js:15
final value = abcde	app.js:16

>

ab password Register



APNA COLLEGE

Elements Console >>

top Filter

Default levels No Issues

final value = abcde	app.js:16
input event	app.js:15
final value = abcdeg	app.js:16
input event	app.js:15
final value = abcdegh	app.js:16
input event	app.js:15
final value = abcdeght	app.js:16
input event	app.js:15
final value = abcdeghti	app.js:16
input event	app.js:15
final value = abcdeght	app.js:16
input event	app.js:15
final value = abcdegh	app.js:16
input event	app.js:15
final value = abcdeg	app.js:16
input event	app.js:15
final value = abcde	app.js:16
input event	app.js:15
final value = abcd	app.js:16
input event	app.js:15
final value = abc	app.js:16
input event	app.js:15

JS

File

/Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

abpasswordRegister

APNA COLLEGE

ElementsConsole

topFilter

Default levelsNo Issues

input eventapp.js:15

final value = aapp.js:16

input eventapp.js:15

final value = abapp.js:16

JS

File

/Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

APNA COLLEGE

My Text Editor

type something

Elements

Console

top

Filter

Default levels

No Issues

<> index.html

style.css

JS app.js

```
JS app.js > inp.addEventListener("input") callback
1 let inp = document.querySelector("#text");
2
3 inp.addEventListener("input", function() {
4   console.log(inp.value)
5 })
```



JS

× +

← → ↻ ⓘ File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

My Text Editor

APNA COLLEGE

Elements Console >>

top Filter

Default levels No Issues

a	app.js:4
ab	app.js:4
abc	app.js:4
abcd	app.js:4

>

<> index.html

style.css

JS app.js X

```
JS app.js > inp.addEventListener("input") callback
1 let inp = document.querySelector("#text");
2 let p = document.querySelector("p");
3
4 inp.addEventListener("input", function () {
5   console.log(inp.value);
6   p.innerText = inp.value;
7 });
8
```



JS

× +

← → ↺

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

My Text Editor

abcdefgh

abcdefgh

APNA COLLEGE

Elements Console >>

top Filter

Default levels No Issues

a	app.js:5
ab	app.js:5
abc	app.js:5
abcd	app.js:5
abcde	app.js:5
abcdef	app.js:5
abcdefg	app.js:5
abcdefgh	app.js:5

>

My Text Editor

shradha khapra is a good student.

shradha khapra is a good student.



APNA COLLEGE

Elements Console >>

top Filter

Default levels No Issues

shradha khapra	app.js:5
shradha khap	app.js:5
shradha khapr	app.js:5
shradha khapra	app.js:5
shradha khapra	app.js:5
shradha khapra i	app.js:5
shradha khapra is	app.js:5
shradha khapra is	app.js:5
shradha khapra is a	app.js:5
shradha khapra is a	app.js:5
shradha khapra is a g	app.js:5
shradha khapra is a go	app.js:5
shradha khapra is a goo	app.js:5
shradha khapra is a good	app.js:5
shradha khapra is a good	app.js:5
shradha khapra is a good s	app.js:5
shradha khapra is a good st	app.js:5
shradha khapra is a good stu	app.js:5
shradha khapra is a good stud	app.js:5
shradha khapra is a good stude	app.js:5
shradha khapra is a good studen	app.js:5
shradha khapra is a good student	app.js:5

JS (Part 10)

Practice Questions

Qs1. Try out the following events in Event Listener on your own :

- mouseout
- keypress
- Scroll
- load

[Use MDN for help]

Qs2. Create a button on the page using JavaScript. Add an event listener to the button that changes the button's color to green when it is clicked.

Qs3. Create an input element on the page with a placeholder "enter your name" and an H2 heading on the page inside HTML.

The purpose of this input element is to enter a user's name so it should only input letters from a-z, A-Z and space (all other characters should not be detected).

Whenever the user inputs their name, their input should be dynamically visible inside the heading.

[Please note that no other character apart from the allowed characters should be visible in the heading]